

SmolVLM: Redefining Small and Efficient Multimodal Models

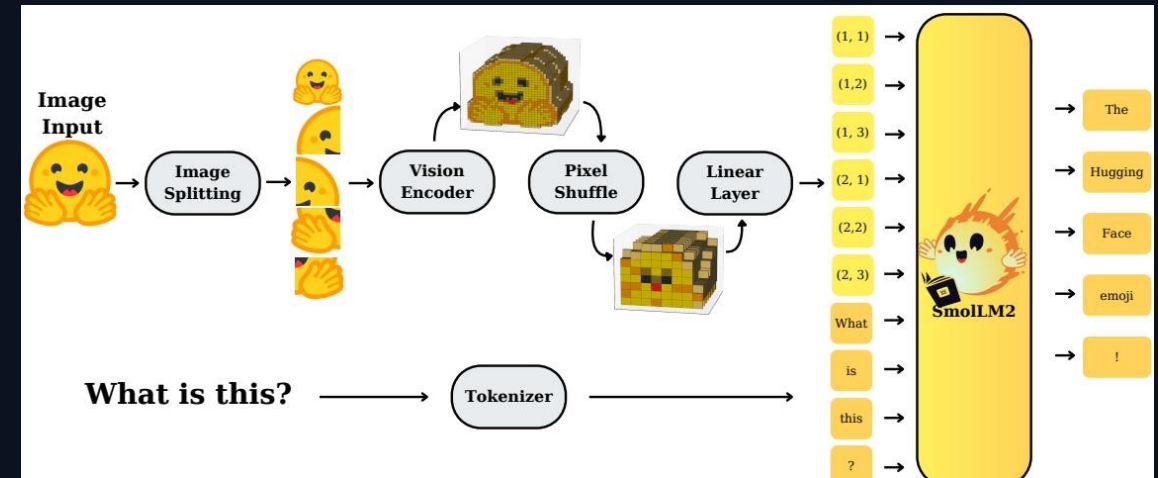
Hugging Face & Stanford University • Compact vision-language models for edge inference

One-liner

SmolVLM is a family of compact VLMs (256M / 500M / 2.2B) designed for resource-efficient on-device inference.

Achieves competitive performance with dramatically lower memory footprints via compute allocation, token compression, and long-context scaling.

All weights, datasets, code, and an iOS app (HuggingSnap) are fully open-sourced.



Large VLMs Are Powerful but Impractical for Edge Deployment

Edge constraints

RAM and bandwidth, not just FLOPs, dominate latency and feasibility on mobile/edge.

State-of-the-art VLMs often require multi-GB memory footprints that exceed consumer devices.

Goal: retain multimodal capability while driving inference into sub-1GB RAM regimes.

Model	Params	RAM (batch=1)
SmoIVLM-256M	256M	0.8 GB
SmoIVLM-500M	500M	1.2 GB
SmoIVLM-2.2B	2.2B	4.9 GB
Molmo-A1B-7B	7B	27.7 GB

SmoIVLM runs with dramatically lower memory while remaining competitive.

Key claim

SmoIVLM-256M runs <1 GB and outperforms the 300× larger Idefics-80B model (per paper).

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM

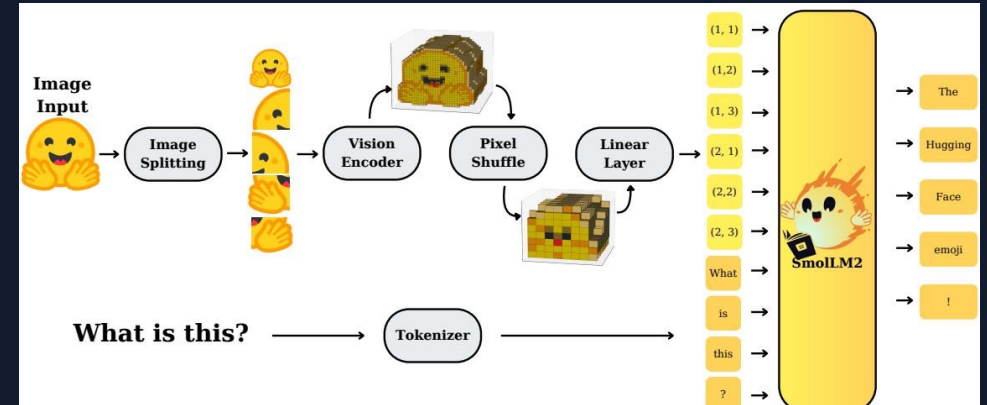
Pipeline (high level)

Split image into sub-images → encode with SigLIP vision encoder.

Pixel shuffle (space-to-depth) compresses spatial tokens.

Linear projection maps vision features into SmolLM2 embedding space.

Concatenate visual tokens + text tokens → autoregressive decoding.



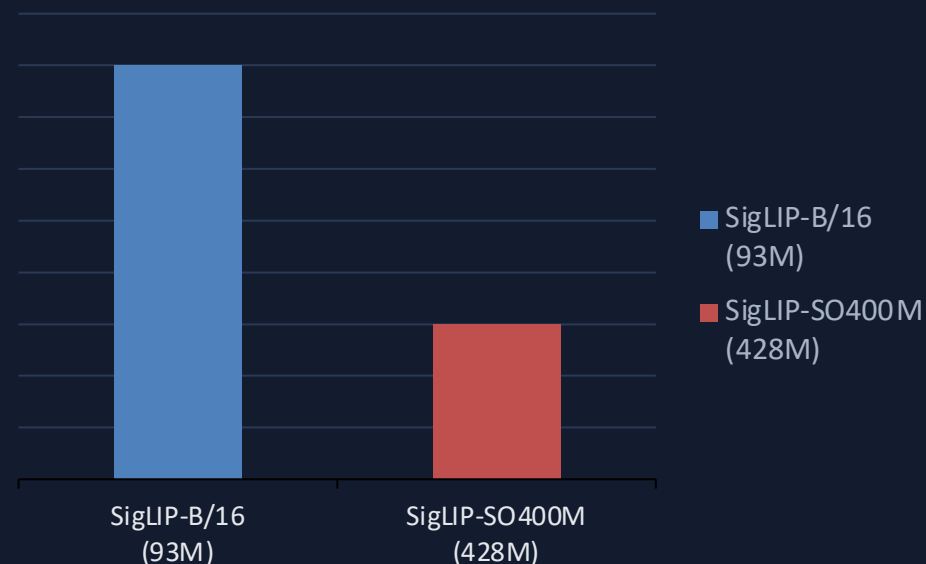
Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1 — balanced encoder/LM

Compact VLMs benefit from smaller vision encoders. With a 135M LM, 93M SigLIP-B/16 outperforms 428M SigLIP-SO400M.

Only at ~1.7B LM scale does the larger encoder justify its ~10% parameter overhead.

Encoder choice for smallest LMs (relative)



Directionality from paper: 93M encoder > 428M encoder for smallest LMs.

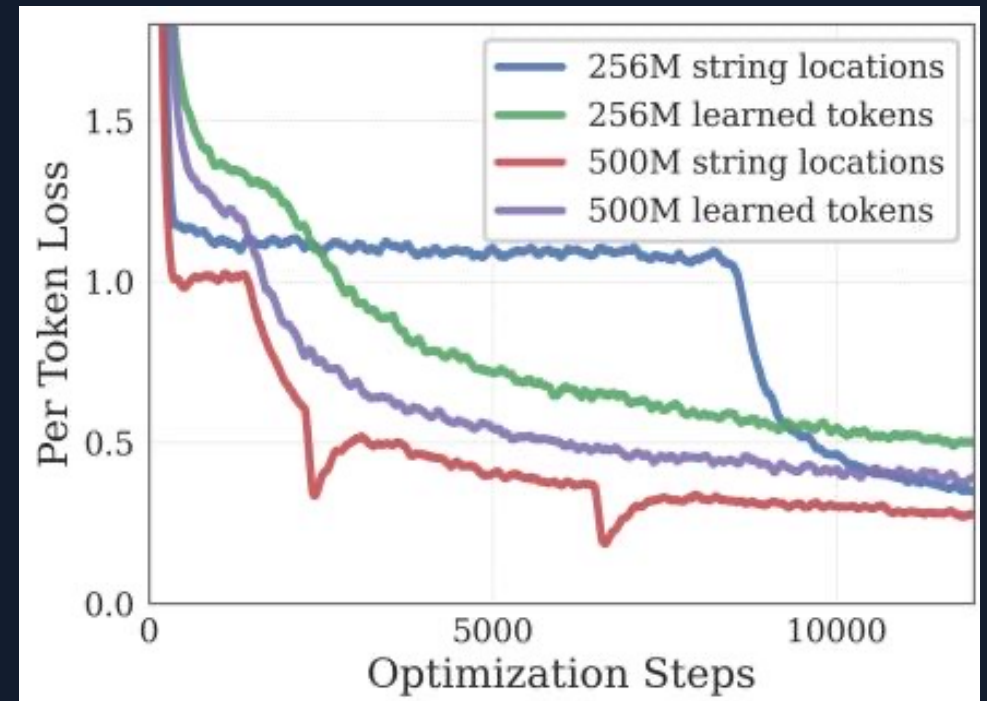
Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small M

Finding 3 — aggressive compression

Pixel shuffle rearranges spatial features into channels, reducing token count by r^2 .

Small models benefit from $r=4$ (16× token reduction); larger models often prefer $r=2$.

Compression reduces attention overhead and improves long-context modeling.



$r=4 \rightarrow 16\times$

Pixel shuffle (space-to-depth) diagram

Extending Context from 2K to 16K Tokens Unlocks Higher Image

Finding 2 — long-context is enabling

Extending context from 2K → 16K tokens significantly improves performance.

Technique: increase RoPE base from 10K → 273K, then fine-tune on mixed long/short-context data.

SmolVLM-2.2B uses 16K context; smaller variants use 8K.

2K (baseline)

8K (256M/500M)

16K (2.2B)

RoPE base: 10K → 273K + mixed-context fine-tuning

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB

Model	Avg (13) ↑	RAM (GB) ↓	Efficient OS baseline
SmolVLM-256M	44.0%	0.8	—
SmolVLM-500M	51.0%	1.2	—
SmolVLM-2.2B	59.8%	4.9	—
Molmo-A1B-7B	—	27.7	Best Efficient OS (single-image/multi-task)
InternVL2-2B	—	—	Best Efficient OS (video)

Notes

SmolVLM evaluated across 13 benchmarks: single-image, multi-task, and video.

SmolVLM-256M runs <1GB RAM (batch=1) and surpasses the 300× larger Ldefics-80B (paper claim).

Real-Time Inference on iPhones: Up to 80 Tokens/Second on App

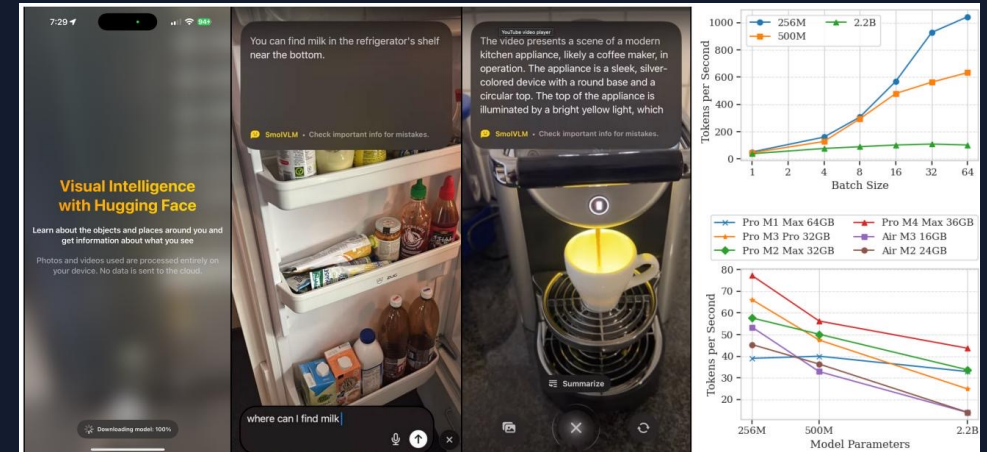
HuggingSnap (on-device)

Runs entirely locally: photos + videos, no cloud dependency.

SmolVLM-256M reaches up to ~80 tokens/s on Apple Silicon (Pro M1 Max 64GB).

On MacBook Air M3 16GB: ~55 tokens/s for the 256M model.

Throughput scales roughly linearly with batch size for the smallest model; at batch=64 approaches ~1000 tokens/s on high-end devices.



HuggingSnap app + throughput plots (paper)

Key Takeaways: Design Principles for Efficient Multimodal Mode

Actionable principles

Balance encoder vs. LM compute; bigger vision backbones can hurt small LMs.

Use aggressive token compression (pixel shuffle $r=4$) when the LM is small.

Scale context length (2K $\rightarrow$ 16K) with RoPE base increase + mixed-context fine-tuning.

Target sub-1GB RAM inference as a first-class design constraint (SmolVLM-256M: 0.8 GB).

Ship reproducible artifacts: weights, datasets, code, and on-device app (fully open-source).

Bottom line: architectural efficiency can dominate raw scale for on-device multimodal systems.